

Hypomorphic homozygous mutations in phosphoglucomutase 3 (*PGM3*) impair immunity and increase serum IgE levels

Atfa Sassi, PhD,^{a*} Sandra Lazaroski, PhD,^{b*} Gang Wu, MSc,^{c*} Stuart M. Haslam, PhD,^c Manfred Fliegauf, PhD,^b Fethi Mellouli, MD,^d Turkan Patiroglu, MD,^{e,f} Ekrem Unal, MD,^e Mehmet Akif Ozdemir, MD,^e Zineb Jouhadi, MD,^g Khadija Khadir, MD,^g Leila Ben-Khemis, MSc,^a Meriem Ben-Ali, PhD,^a Imen Ben-Mustapha, MD,^a Lamia Borchani, PhD,^h Dietmar Pfeifer, PhD,ⁱ Thilo Jakob, MD,^j Monia Khemiri, MD,^k A. Charlotta Asplund, BSc,^l Manuela O. Gustafsson, MSc,^l Karin E. Lundin, PhD,^l Elin Falk-Sörqvist, MSc,^m Lotte N. Moens, PhD,^m Hatice Eke Gungor, MD,^f Karin R. Engelhardt, PhD,ⁿ Magdalena Dziadzio, MD, PhD,ⁿ Hans Stauss, MD, PhD,ⁿ Bernhard Fleckenstein, MD,^o Rebecca Meier, BSc,^b Khairunnadiya Prayitno, MSc,^b Andrea Maul-Pavacic, PhD,^b Sandra Schaffer,^b Mirzokhid Rakhmanov, PhD,^b Philipp Henneke, MD,^b Helene Kraus, MSc,^b Hermann Eibel, PhD,^b Uwe Kölsch, MD,^p Sellama Nadifi, PhD,^q Mats Nilsson, PhD,^m Mohamed Bejaoui, MD,^d Alejandro A. Schäffer, PhD,^{r*} C. I. Edvard Smith, MD, PhD,^{l*} Anne Dell, PhD,^{c*} Mohamed-Ridha Barbouche, MD, PhD,^{a*} and Bodo Grimbacher, MD^{b,n*}

Tunis, Tunisia, Freiburg, Erlangen, and Berlin, Germany, London, United Kingdom, Kayseri, Turkey, Casablanca, Morocco, Huddinge and Uppsala, Sweden, and Bethesda, Md

Background: Recurrent bacterial and fungal infections, eczema, and increased serum IgE levels characterize patients with the hyper-IgE syndrome (HIES). Known genetic causes for HIES are mutations in signal transducer and activator of transcription 3 (*STAT3*) and dedicator of cytokinesis 8 (*DOCK8*), which are involved in signal transduction pathways. However, glycosylation defects have not been described in patients with HIES. One crucial enzyme in the glycosylation pathway is phosphoglucomutase 3 (*PGM3*), which catalyzes a key step in the synthesis of uridine diphosphate N-acetylglucosamine, which is required for the biosynthesis of N-glycans.

Objective: We sought to elucidate the genetic cause in patients with HIES who do not carry mutations in *STAT3* or *DOCK8*. **Methods:** After establishing a linkage interval by means of SNPchip genotyping and homozygosity mapping in 2 families with HIES from Tunisia, mutational analysis was performed with selector-based, high-throughput sequencing. Protein expression was analyzed by means of Western blotting, and glycosylation was profiled by using mass spectrometry. **Results:** Mutational analysis of candidate genes in an 11.9-Mb linkage region on chromosome 6 shared by 2 multiplex families identified 2 homozygous mutations in *PGM3* that segregated

From ^athe Laboratory of Immunopathology, Vaccinology and Molecular Genetics, Pasteur Institute of Tunis and University Tunis El Manar; ^bthe Center for Chronic Immunodeficiency (CCI), ^cthe Department of Medicine I, Specialties: Hematology, Oncology, and Stem-Cell Transplantation, and ^dthe Allergy Research Group, Department of Dermatology, University Medical Center Freiburg; ^ethe Department of Life Sciences, Imperial College London; ^fthe Pediatrics Department, Bone Marrow Transplantation Center, Tunis; ^gthe Department of Pediatrics, Division of Pediatric Hematology and Oncology, and ^hthe Department of Pediatrics, Division of Pediatric Immunology, Faculty of Medicine, Erciyes University, Kayseri; ⁱthe Department of Pediatric Infectious Diseases, CHU IBN ROCHD, and ^jthe Department of Genetics, Hassan II University, Casablanca; ^kthe Laboratory of Venoms and Therapeutic Molecules, Institut Pasteur de Tunis; ^lPediatrics Department A, Children's Hospital of Tunis; ^mthe Clinical Research Center, Department of Laboratory Medicine, Karolinska Institutet, Karolinska University Hospital Huddinge; ⁿthe Department of Immunology, Genetics and Pathology, Uppsala University, Rudbeck Laboratory, Uppsala; ^othe Royal Free Hospital, Institute of Immunity & Transplantation, University College London; ^pthe Institute of Virology, Friedrich-Alexander-Universität Erlangen-Nürnberg, Erlangen; ^qthe Division of Immunology, Labor Berlin and Institute of Medical Immunology, Charité, Campus Virchow Klinikum, Berlin; and ^rthe National Center for Biotechnology Information, National Institutes of Health, Department of Health and Human Services, Bethesda.

*These authors contributed equally to this work.

Supported by the German Federal Ministry of Education and Research (BMBF 01 EO 0803). The research was supported in part by the Intramural Research Program of the National Library of Medicine–National Institutes of Health. Parts of the study were supported by the Tunisian Ministry for Higher Education and Research, the Swedish Medical Research Council, the Swedish Cancer Society and the Stockholm County Council (research grant ALF), and the European Community's 6th and 7th Framework Programs FP7/2007-2013 under grant agreement

Health-F5-2008-223292 (Euro-Gene-Scan). This work was also supported by the Biotechnology and Biological Sciences Research Council (BBF0083091 and BB/K016164/1 to A.D. and S.M.H.). G.W. is grateful for a Sun Hung Kai Properties-Kwoks' Foundation-Imperial College PhD Scholarship.

Disclosure of potential conflict of interest: S. M. Haslam and A. Dell have received research support from the Biotechnology and Biological Sciences Research Council. L. Ben-Khemis, M. Ben-Ali, I. Ben-Mustapha, M. Khemiri, and M.-R. Barbouche have received research support from the Tunisian Ministry for Higher Education and Research. T. Jakob has consultant arrangements with Allergopharma, Novartis, Jansen Cilag, and Thermo Fisher Scientific; receives research support from Allergopharma, Thermo Fisher Scientific, Birken AG, and Cosmetic Europe; and has received payment for lectures from Stallergenes, ALK-Abelló, Allergies Therapeutics, Novartis, and Thermo Fisher Scientific. A. C. Asplund, M. O. Gustafsson, K. E. Lundin, and C. I. E. Smith have received research support and travel support from the European Union Framework Programme 7. H. Stauss has received research support from Leukemia & Lymphoma Research, has consultant arrangements with Cell Medica, has a patent for Allo-restricted cells, and receives royalties for that patent. B. Fleckenstein has received research support from DFG. B. Grimbacher has received research support from the BMBF, the European Union, and the Helmholtz Society; and has received payment for lectures from CSL, Baxter, and Biotest. The rest of the authors declare that they have no relevant conflicts of interest.

Received for publication July 1, 2013; revised January 31, 2014; accepted for publication February 4, 2014.

Available online April 7, 2014.

Corresponding author: Bodo Grimbacher, MD, Center for Chronic Immunodeficiency (CCI), University Medical Center Freiburg, Engesserstraße 4, 79108 Freiburg, Germany. E-mail: bodo.grimbacher@uniklinik-freiburg.de.

0091-6749

<http://dx.doi.org/10.1016/j.jaci.2014.02.025>

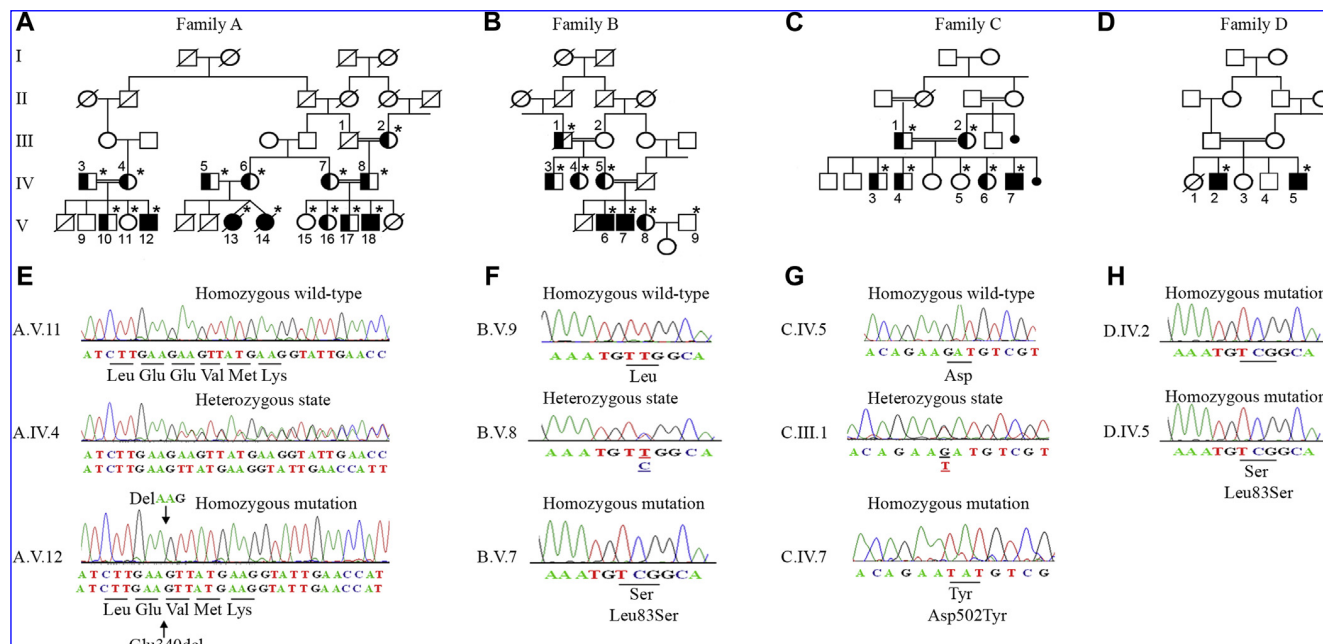


FIG 1. Segregation of *PGM3* mutations with disease status in families with AR-HIES. **A-D**, Family A, p.Glu340del; families B and D, p.Leu83Ser; and family C, p.Asp502Tyr. Circles, Female subjects; squares, male subjects; solid symbols, affected subjects with homozygous mutations; half-solid symbols, heterozygous carriers; open symbols, healthy members with wild-type *PGM3*; slashes, deceased subjects; double horizontal lines, consanguinity; black dots, miscarriages. Asterisks indicate mutations confirmed by using sequencing. **E-H**, Sequence analyses of unaffected subjects without *PGM3* mutations (top), unaffected heterozygous carriers (middle in Fig 1, E-G), and homozygous affected subjects (bottom).

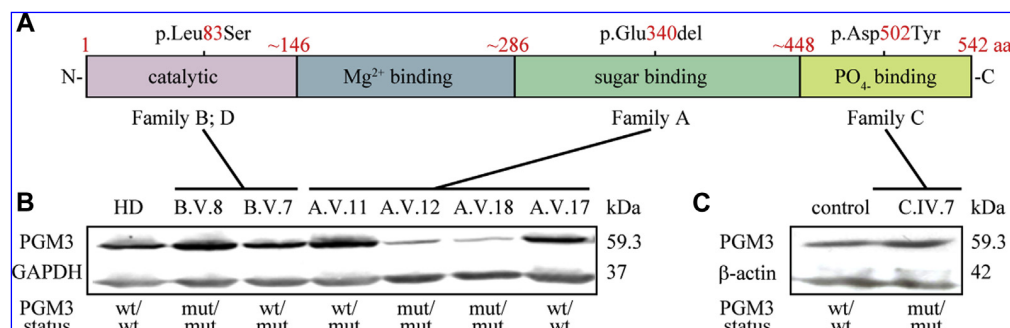


FIG 2. Homozygous p.Glu340del mutations within the sugar-binding domain cause reduced *PGM3* expression. **A**, Identified mutations localize to distinct protein domains. Numbers indicate amino acid residues. **B**, Western blotting shows reduced *PGM3* abundance in EBV-transformed B cells from both homozygous carriers of family A. Mutational status is indicated below each lane. Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) confirms equal loading. HD, Healthy donor. **C**, Normal *PGM3* expression in immortalized T cells from the patient in family C who had insufficient B-cell numbers. β -actin was used as a loading control.

patients (A.V.12 and C.IV.7), did not affect the glycosylation pattern of transferrin in the blood (see Fig E3 in this article's Online Repository at www.jacionline.org), which is a commonly used diagnostic test for congenital defects of glycosylation.²⁵ Enzymatic testing revealed that the catalytic activity is reduced in mutant *PGM3*, depending on the substrate dose, with 20% to 30% of residual activity (Fig 3, B). The effect was more pronounced with the p.Glu340del mutation (family A; $P < .0005$ at 12.5 μ mol/L GlcNAc-6-P) than with the p.Leu83Ser mutation (family B, $P < .002$). We concluded that the mutant *PGM3* retains

its catalytic specificity, although with impaired enzymatic activity. In the accompanying article by Zhang et al,²⁶ a biochemical defect was shown by analyzing intracellular UDP-GlcNAc levels.

PGM3 mutations inhibit the formation of complex N-glycan subtypes

By using a common substrate (UDP-GlcNAc), the GlcNAc transferases I (Michaelis-Menten constant K_m 0.04) and II

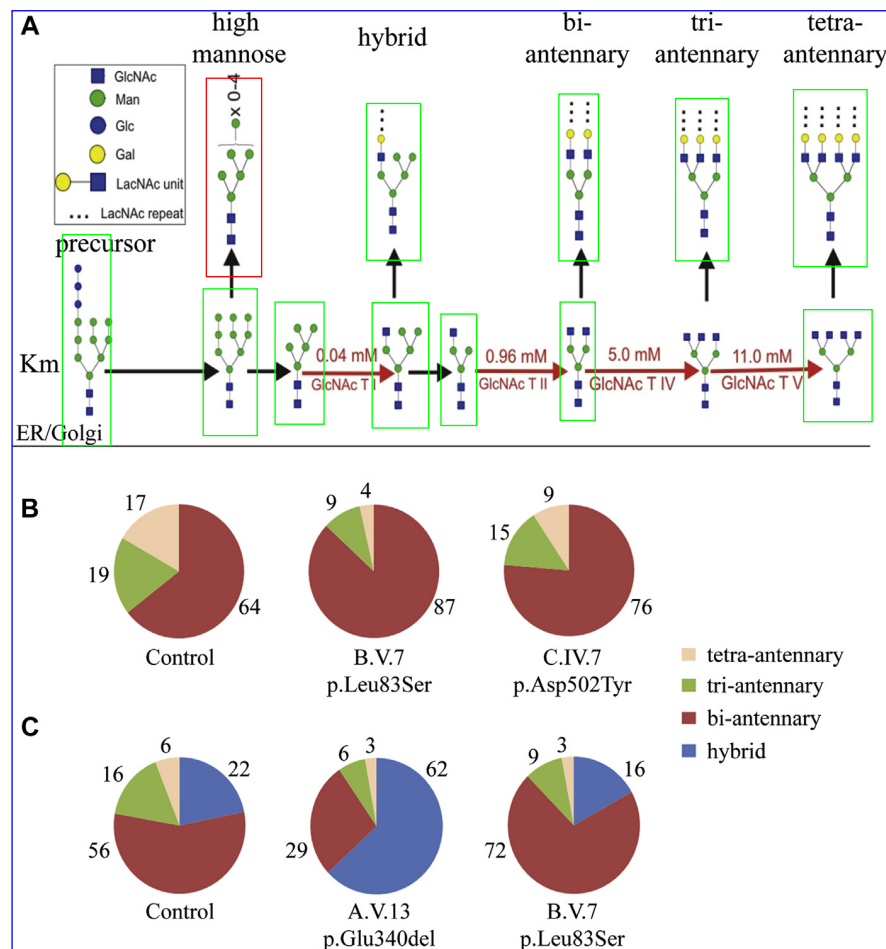


FIG 4. Incomplete assembly of complex-type glycans in patients with *PGM3* mutations. **A**, *PGM3* links to N-glycosylation through UDP-GlcNAc (red). The common substrate for assembly of the complex-type N-glycans in the Golgi apparatus (vertical arrows) is used by 4 GlcNAc transferases (transferase [T] I, II, IV, and V; red arrows) with decreasing catalytic activity (increasing Km values [mmol/L UDP-GlcNAc]).²⁷ UDP-GlcNAc produced by mutant *PGM3* might be insufficient to allow for completion of N-glycan antenna formation. ER, Endoplasmic reticulum. **B**, Accumulation of bi-antennary N-glycans in neutrophils²⁸ from patients with *PGM3* mutations. Decreased relative abundance of tri-antennary and tetra-antennary glycans (percentage values of N-glycans detected by using glycomic profiling; see Figs E4 and E5) in affected patients B.V.7 (p.Leu83Ser; catalytic domain) and to a lesser extent in affected patient C.IV.7 (p.Asp502Tyr; phosphate-binding domain). **C**, The homozygous *PGM3* mutation p.Glu340del causes accumulation of the hybrid-type N-glycans, with a simultaneous decrease in bi-antennary, tri-antennary, and tetra-antennary N-glycan levels in EBV B cells from patient A.V.13 (p.Glu340del; sugar-binding domain). B cells from patient B.V.7 (p.Leu83Ser; catalytic domain) have decreased tri-antennary and tetra-antennary N-glycan levels and increased bi-antennary N-glycan levels. Percentage values indicate the relative abundance of glycan types.

Impaired T-cell proliferation, T_H2 skewing, and borderline T_H17 cell numbers in patients with *PGM3* mutations

To test whether the HIES phenotype in our patients with *PGM3* mutations is associated with impaired T-cell function, we analyzed freshly isolated PBMCs using fluorescence-activated cell sorting (FACS) analysis for proliferation and T cell-specific cytokine responses after stimulation *ex vivo* (Tables III and IV and see Figs E1 and E2; see also the accompanying article by Zhang et al²⁶). T-cell proliferation to recall antigens (purified protein derivative of tuberculin and tetanus toxoid) was reduced in all samples tested (Table IV). Interestingly, T-cell proliferation after stimulation with PHA or anti-CD3 of family B carrying the missense mutation (p.Leu83Ser) was comparable with that seen

in healthy control subjects; however, proliferation in family A, with the more severe mutation (p.Glu340del), was again borderline low.

The serum IgE level increase might be explained by abnormal T_H cell differentiation. Therefore we analyzed T_H1-T_H2 subsets in our patients by using intracellular IL-4 and IFN- γ staining (Table III and see Fig E1). Two of 4 patients tested had grossly increased percentages of T_H2 cells (19% and 27%, respectively), likely contributing to the increased serum IgE levels seen in these patients.

In contrast to the other patients, B cells were absent in patient C.IV.7, and an unusual subpopulation of CD4⁺ cells was detected, simultaneously expressing T_H1 and T_H2 cytokines, as well as a reduced proportion of forkhead box protein 3-expressing regulatory T cells (data not shown).

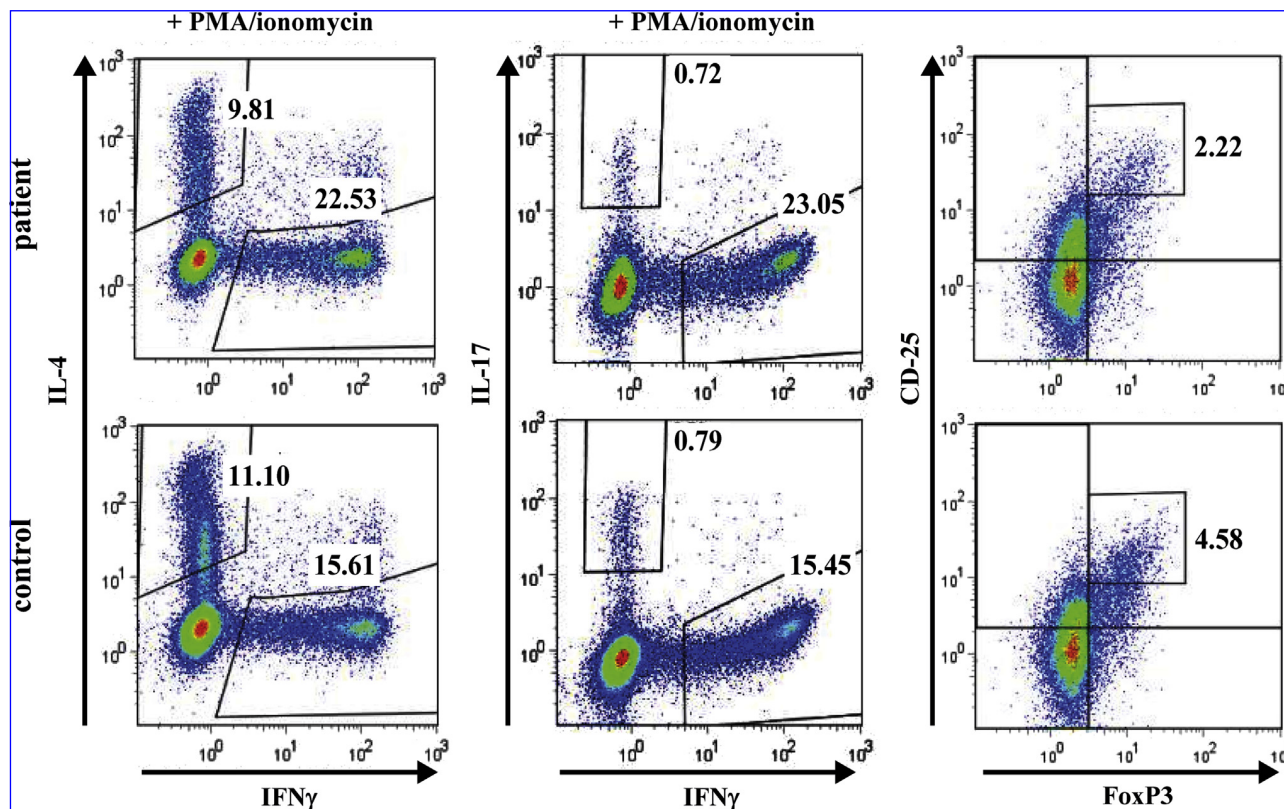


FIG E1. T_H2 skewing but normal T_H17 cell numbers in a patient with *PGM3* mutations. Freshly isolated PBMCs from patient B.V.6 were stimulated, and cytokine profiles were determined by using FACS analysis, as indicated. *Left panels:* upper left gate, IL-4-producing T_H2 cells; lower right gate, IFN- γ -producing T_H1 cells. *Middle panels:* upper left gate, IL-17-producing T_H17 cells; lower right gate, IFN- γ -producing T_H1 cells. *Right panels:* Gate shows the proportion of CD25^{high}/forkhead box protein 3 (*FoxP3*)⁺ regulatory T cells.

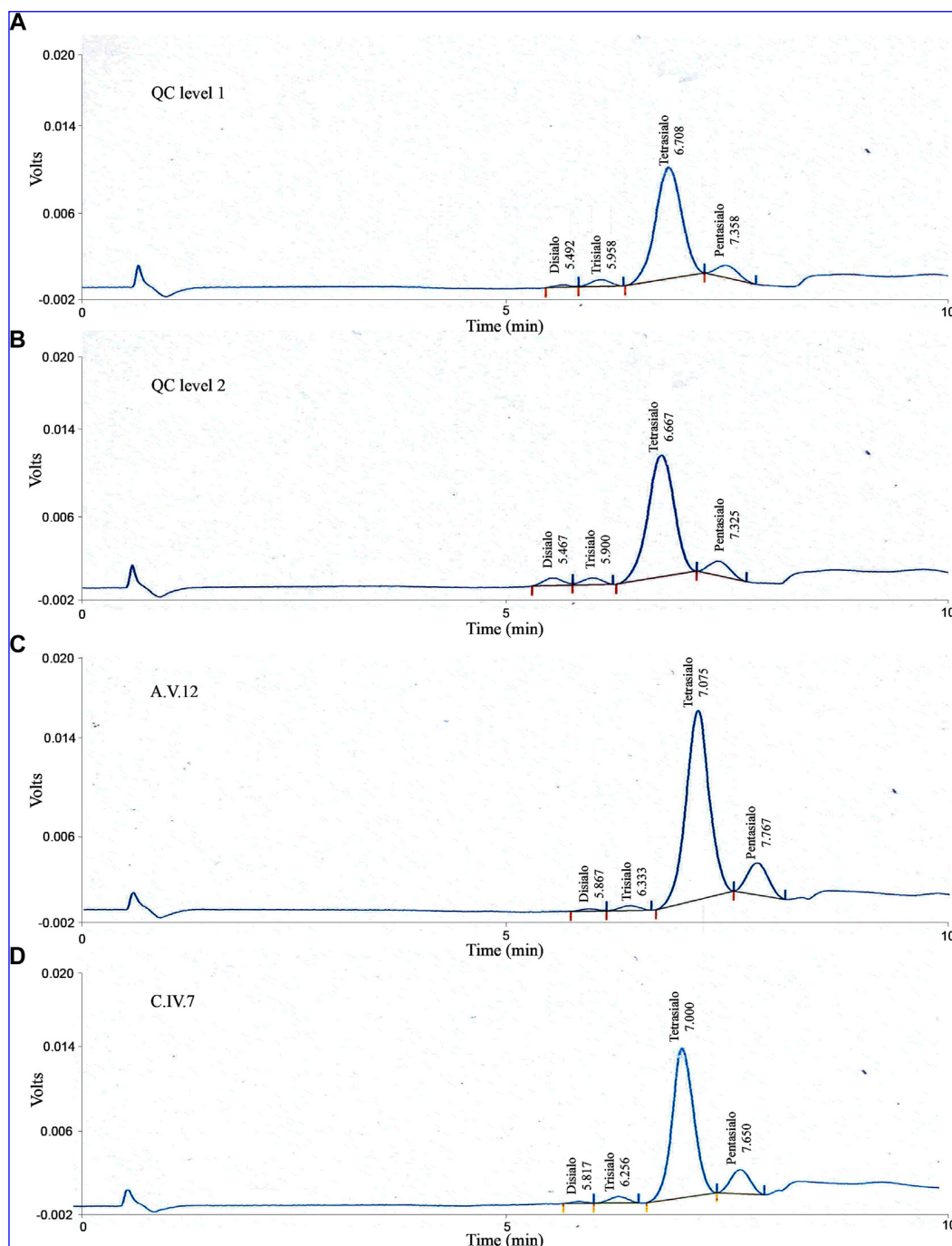


FIG E3. Normal transferrin glycoform patterns in sera from patients with *PGM3* mutations. **A** and **B**, Asialo-, mono-, di-, tri-, tetra-, and penta-sialo-transferrin glycoform patterns in a control serum were analyzed by using HPLC (quality control [QC] level 1 and 2). **C** and **D**, Normal HPLC chromatograms demonstrate normal relative distribution of transferrin glycoforms in sera from patients carrying *PGM3* mutations: A.V.12 (p.Glu340del) and C.IV.7 (p.Asp502Tyr). Transferrin carries 2 bi-antennary N-linked glycans with terminal sialic acid residues, which are used for diagnostic testing for CDGs. Defective transferrin glycosylation would be indicated by an increase of at least one of the asialotransferrin, mono-sialotransferrin, di-sialotransferrin, or tri-sialotransferrin glycoform levels and a decrease in tetra-sialotransferrin levels.

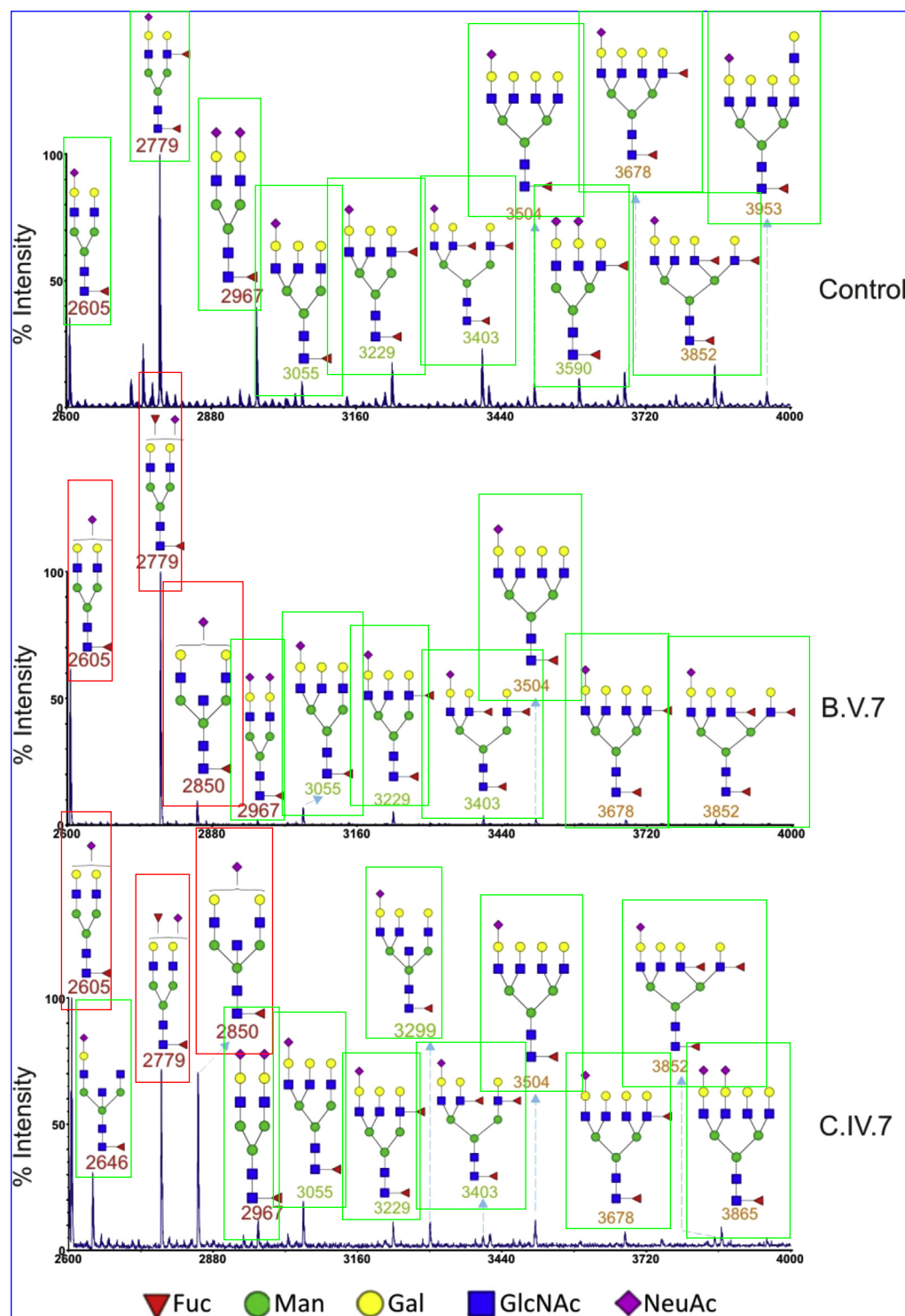


FIG E4. Accumulation of bi-antennary N-glycans in neutrophils from patients with *PGM3* mutations. Mass spectra of permethylated N-glycans of neutrophils from a healthy control subject and the affected subjects B.V.7 and C.IV.7. Results are summarized in Fig 4, B, with consistent color codes. Glycomic data of the control is from Babu et al.^{E16}